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A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere. The Earth is a vibrant blue and green sphere against a dark, star-filled background. A thin white rectangular border is superimposed over the center of the Earth.

COMP-2032
**Introduction to
Image Processing**

Lecture 9
CNN



Learning Outcomes

IDENTIFY

1. Background
2. Components
3. Applications
4. Future





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Background



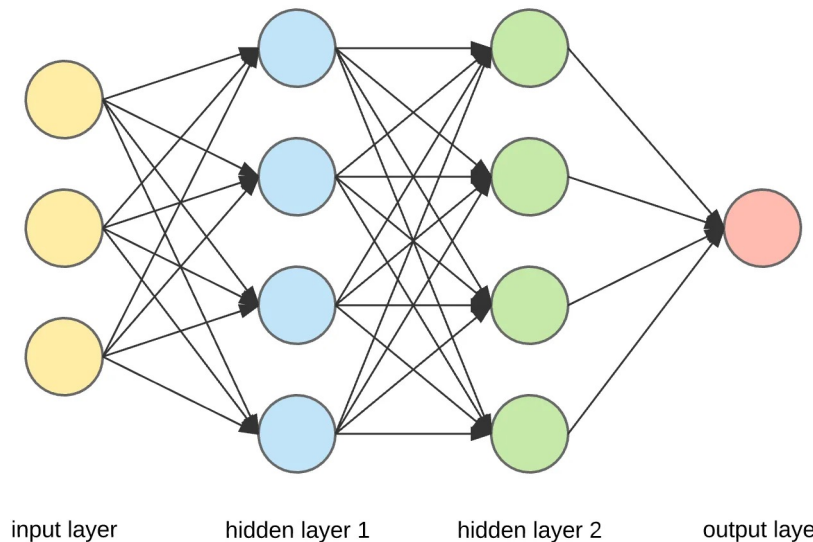
Inspiration...

CNN

Convolutional Neural Network

DID YOU KNOW: Each of the convolutional layers perform the image processing techniques that we learned throughout this module. Techniques such as *convolution/filtering*, *re-sizing*, *noise removal* and *edge detection* to name a few.

Interesting FACT



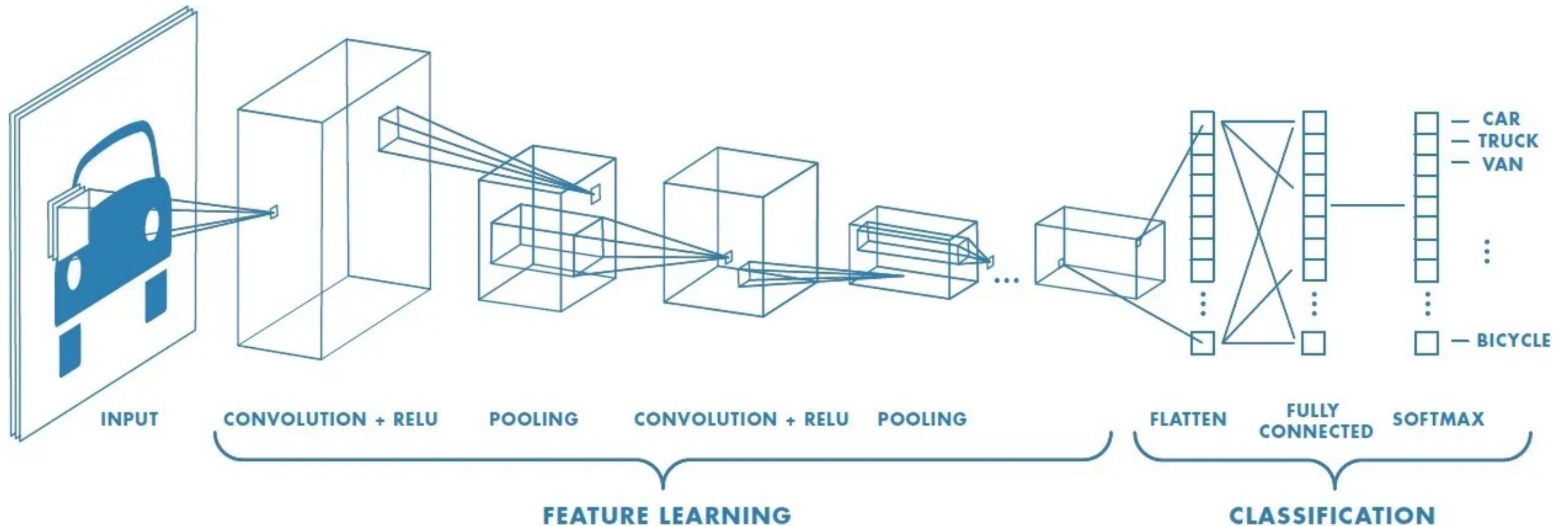
Artificial Neural Network¹

Inspired by

¹ Dertat, A. (8 August 2017). *Applied Deep Learning – Part 1: Artificial Neural Network*. Towards Data Science. <https://towardsdatascience.com/applied-deep-learning-part-1-artificial-neural-networks-d7834f67a4f6>



Convolutional Neural Network **Architecture**



Extracted from: Raghav, P. (4 May 2018). *Understanding of Convolutional Neural Network (CNN) – Deep Learning*. Medium. <https://medium.com/@RaghavPrabhu/understanding-of-convolutional-neural-network-cnn-deep-learning-99760835f148>



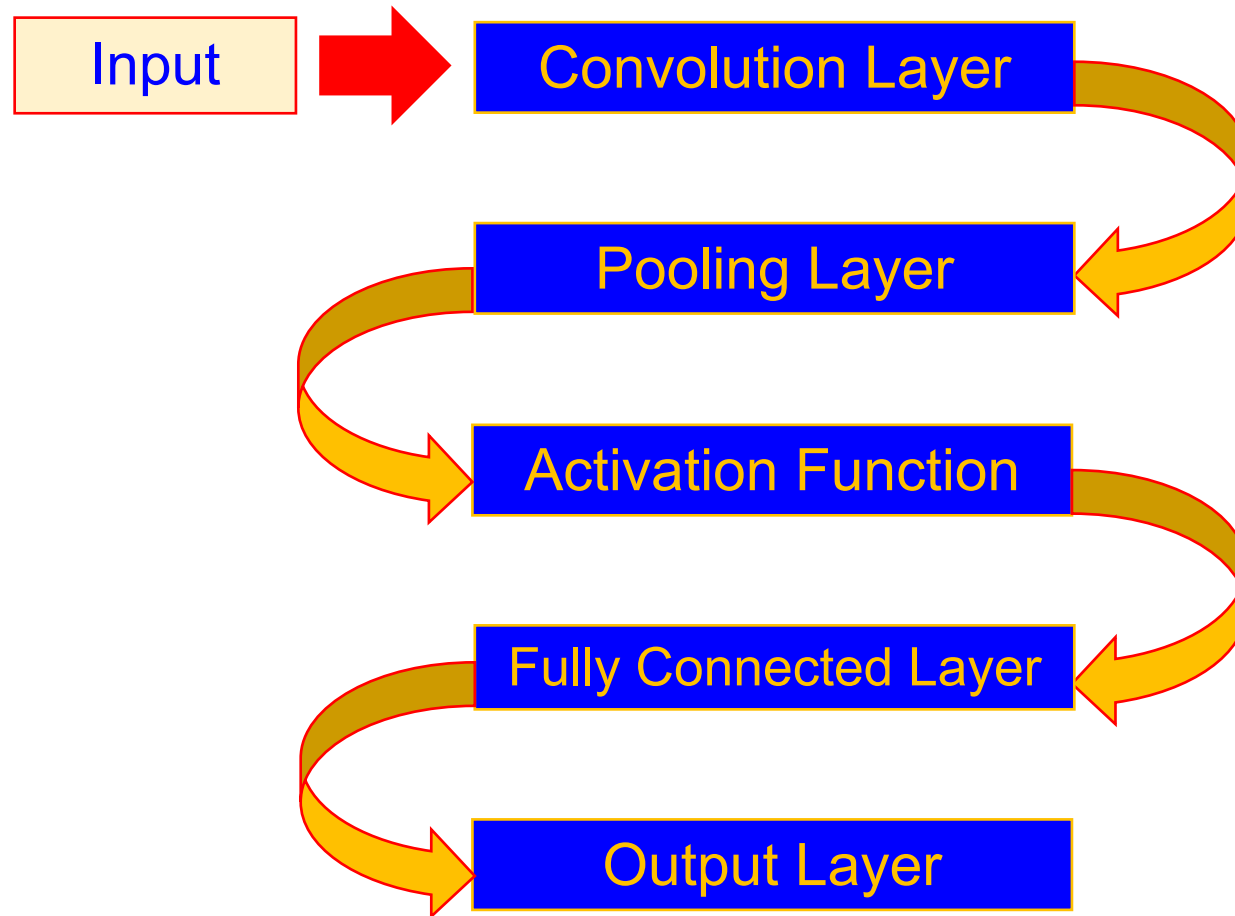
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Components



CNN Components

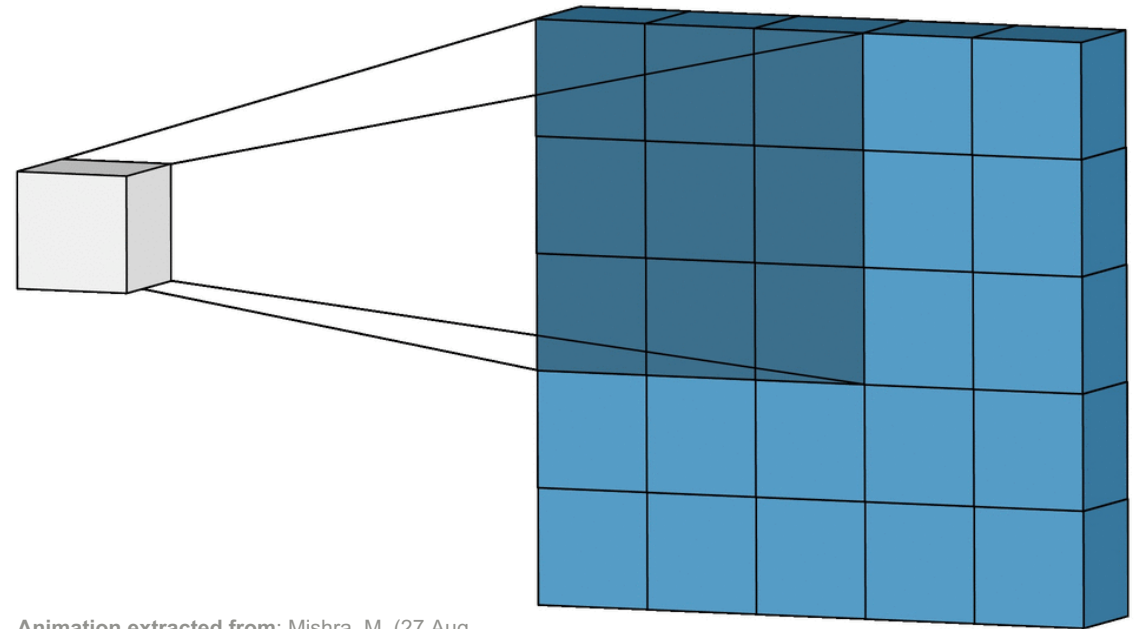




Convolutional Layer

- Most important layer
- Performs major computations
- Takes in **input image**, performs **filtering** which produces **feature map(s)**
- Filters using image processing techniques (e.g., **edge detection**, **blur** and **sharpen**)
- Filtering is performed using 3x3 kernels to perform the dot product

Resulting array [with weights] (*shown in grey in the animation below*) is known as **feature map** or **activation map**



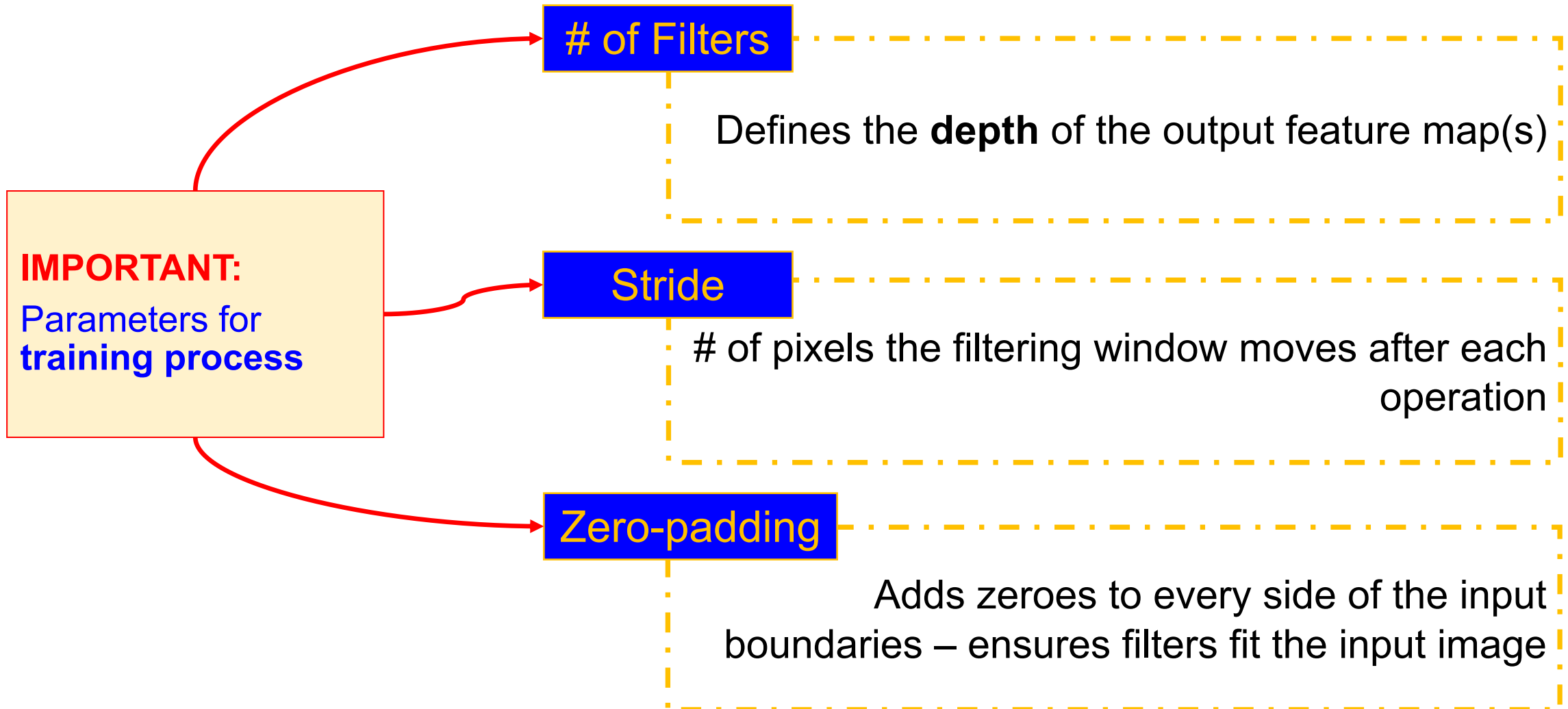
Animation extracted from: Mishra, M. (27 Aug 2020). *Convolutional Neural Networks, Explained*. Medium.
<https://towardsdatascience.com/convolutional-neural-networks-explained-9cc5188c4939>

Filter weights are only **adjusted** via backpropagation & gradient decent during the training process

Note



Convolutional Layer





Convolutional Layer

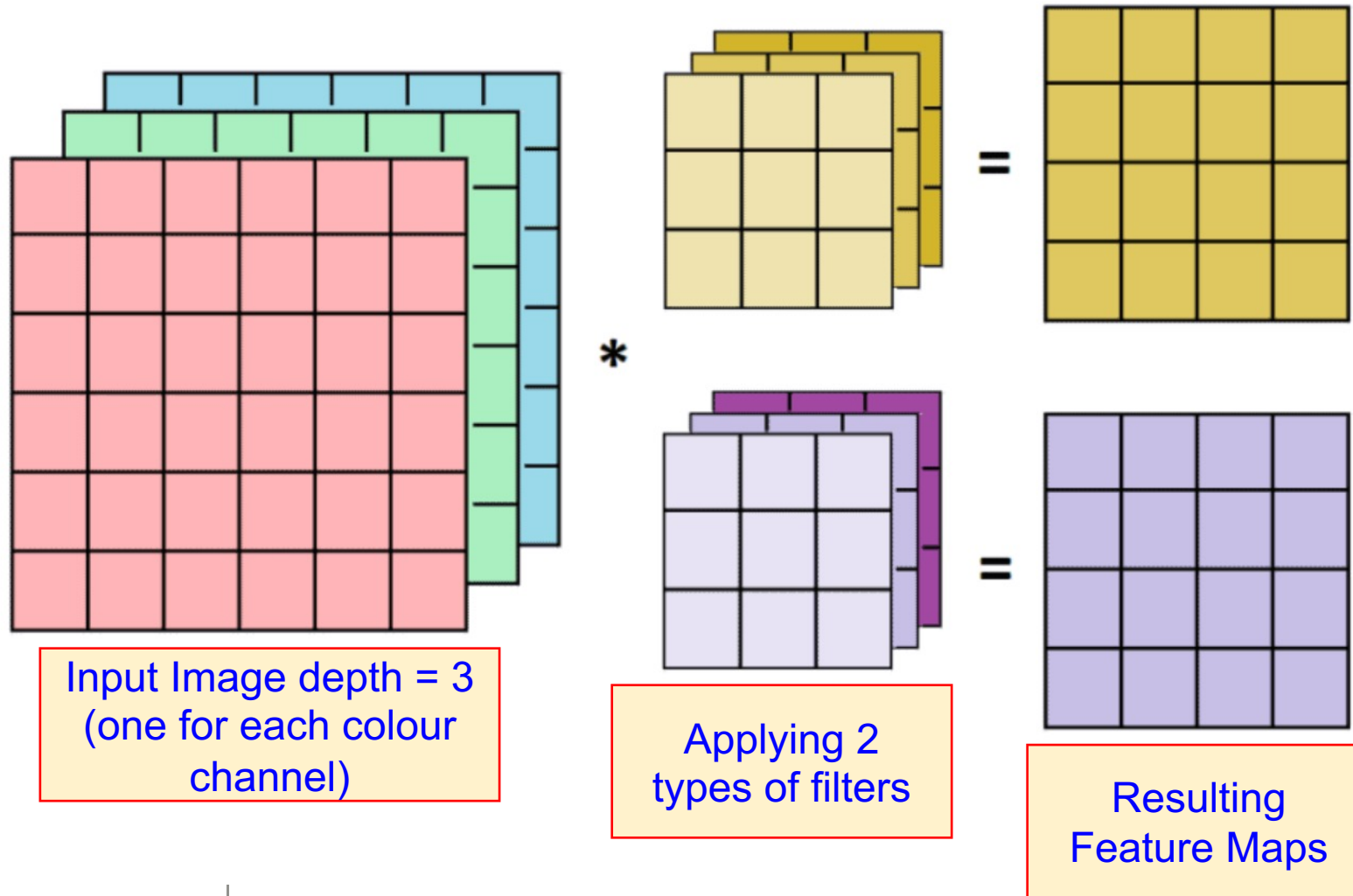


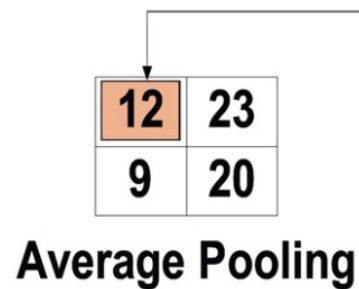
Illustration extracted from: Zvornicanin, E. (18 March 2024). *What is Depth in Convolutional Neural Network?*
<https://www.baeldung.com/cs/cnn-depth>



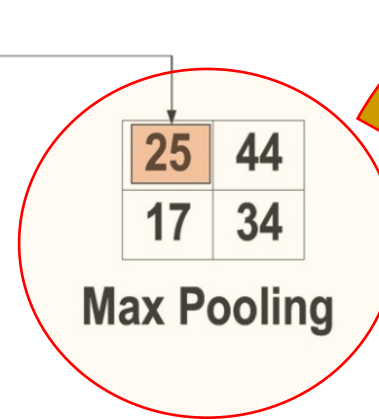
Pooling Layer



Occurs after a convolutional layer. Reduces the dimensionality of the resulting conv layer

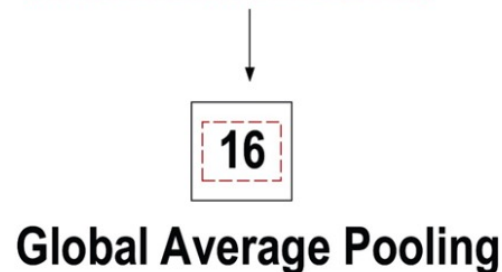


4	25	44	10
8	14	8	33
17	2	16	34
5	13	24	7



Famously Used

Three Types of Pooling Operations



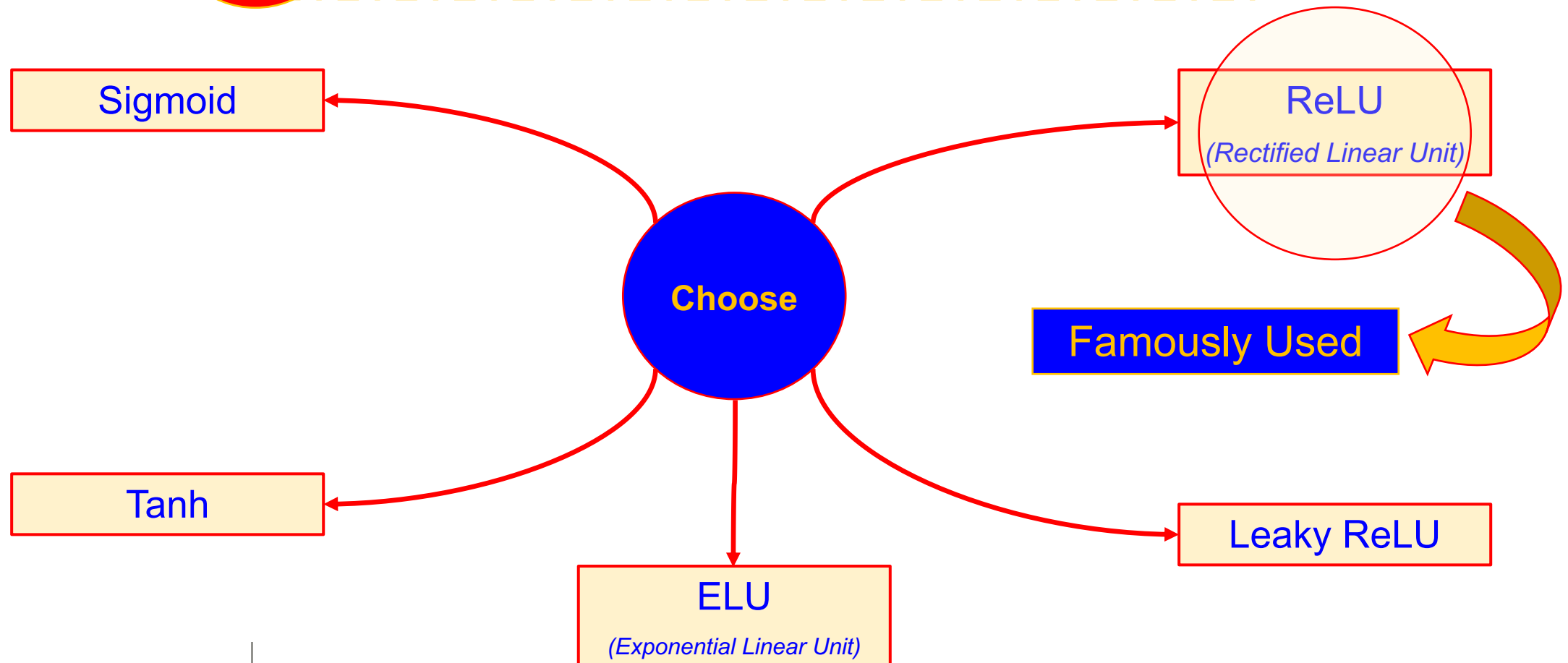
Extracted from: Alzubaidi, L., Zhang, J., Humaidi, A.J., Al-Dujaili, A., Duan, Y., Al-Shamma, O., Santamaría, J., Fadhel, M.A., Al-Amidie, M. and Farhan, L. (2021). Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. *Journal of big Data*, 8, pp.1-74



Activation Function

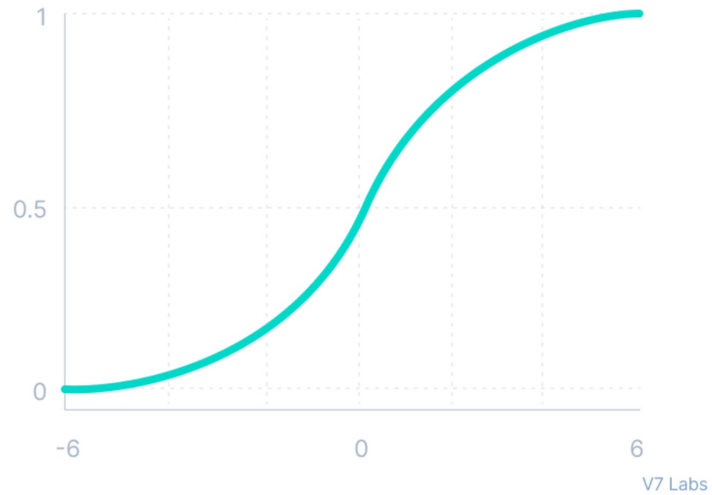
?

A.k.a. *transfer function* – determines the neural network's output

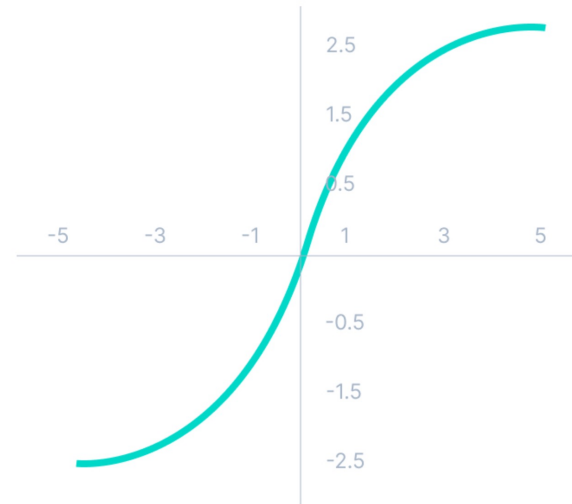




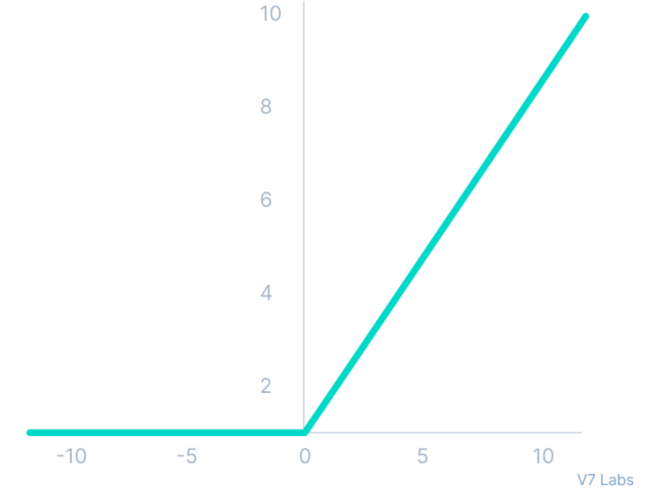
Sigmoid



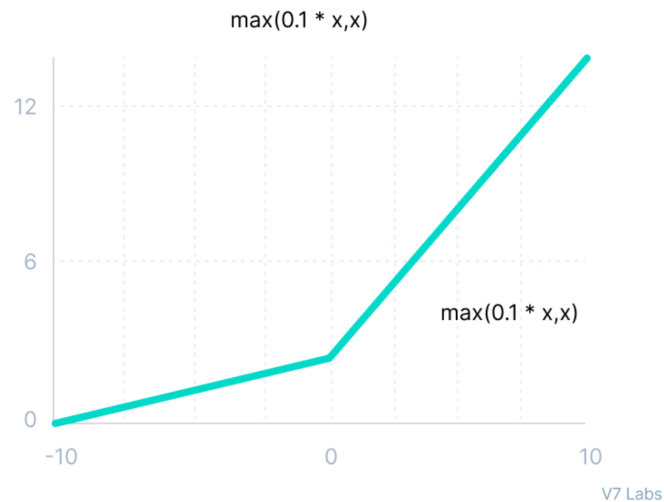
Tanh



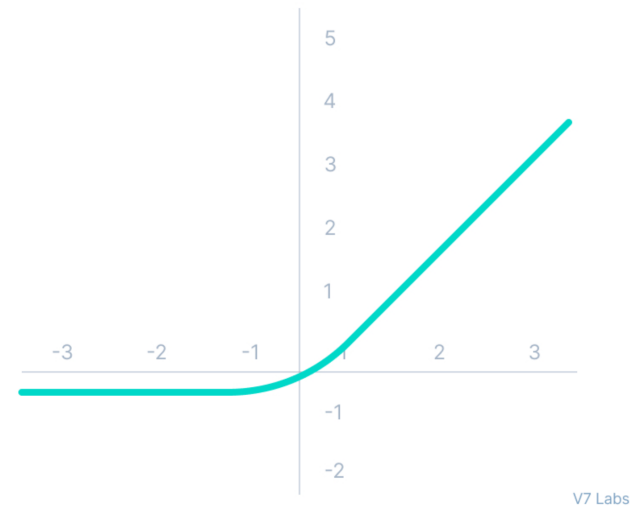
ReLU



Leaky ReLU



ELU



Graphs extracted from: Bagheti, P. (27 May 2021). *Activation Functions in Neural Networks [12 Types & Use Cases]*. V7.
<https://www.v7labs.com/blog/neural-networks-activation-functions>



Fully Connected Layer

- **FINALE** layer
- Utilises features extracted from previous layers, performs task classification

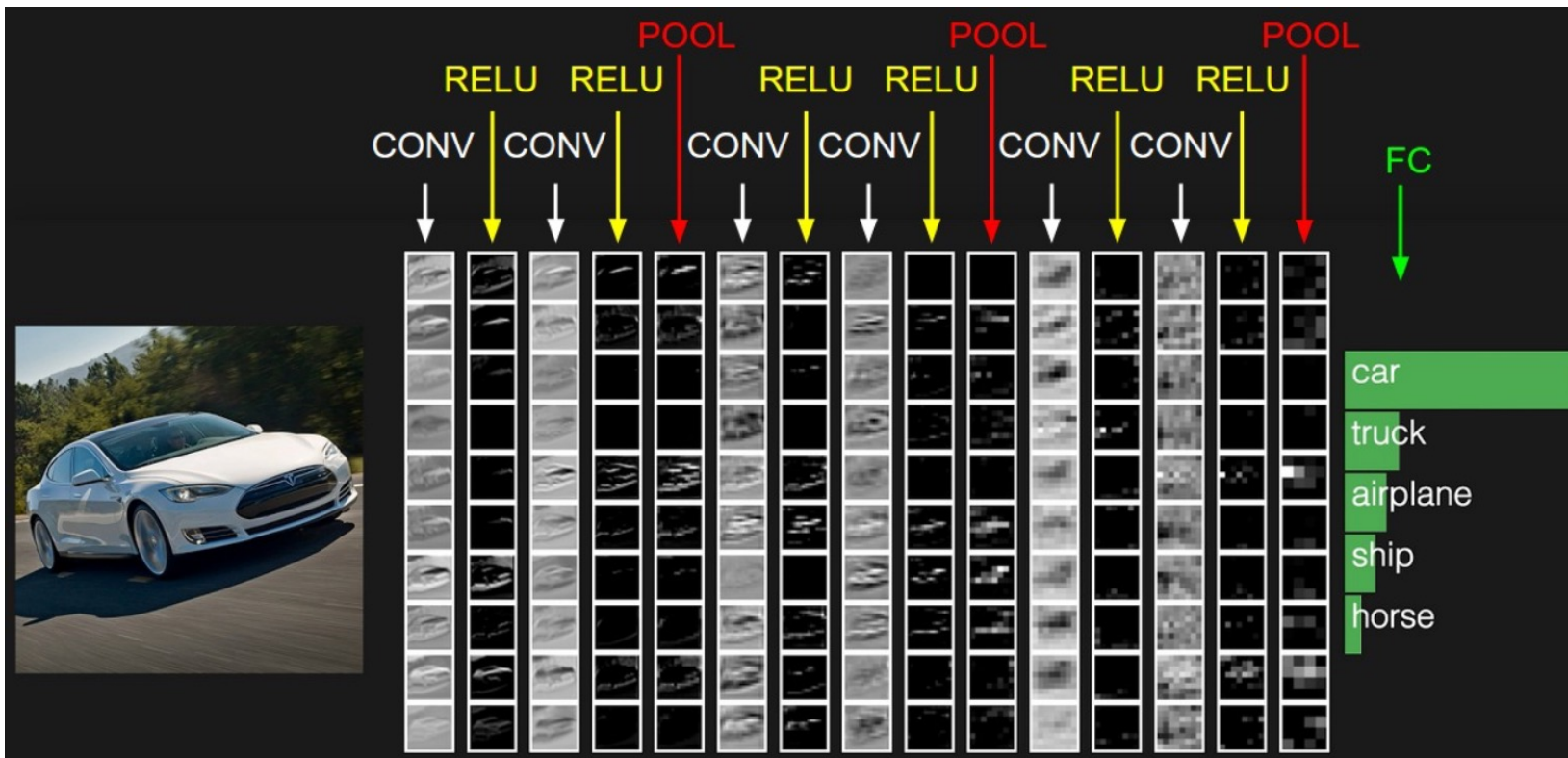


Image extracted from: Standard University.
(n.d.). CS231n Convolutional Neural Networks for
Visual Recognition.
<https://cs231n.github.io/convolutional-networks/#fc>

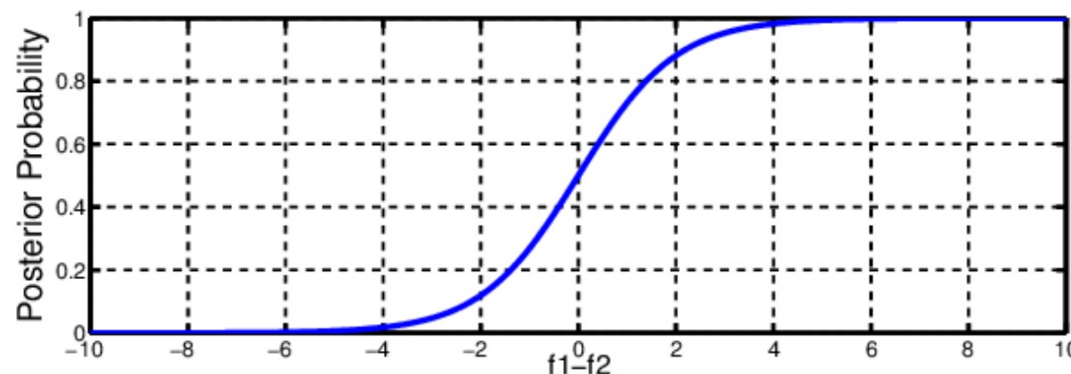
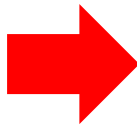


Output Layer

?

- Performs a *logistic function* to classify tasks
- Uses **Softmax** activation function where probability ranges from 0 to 1 – scores given to each class
- Sometimes, **embedded** within the FC layer

Softmax
Activation
Function



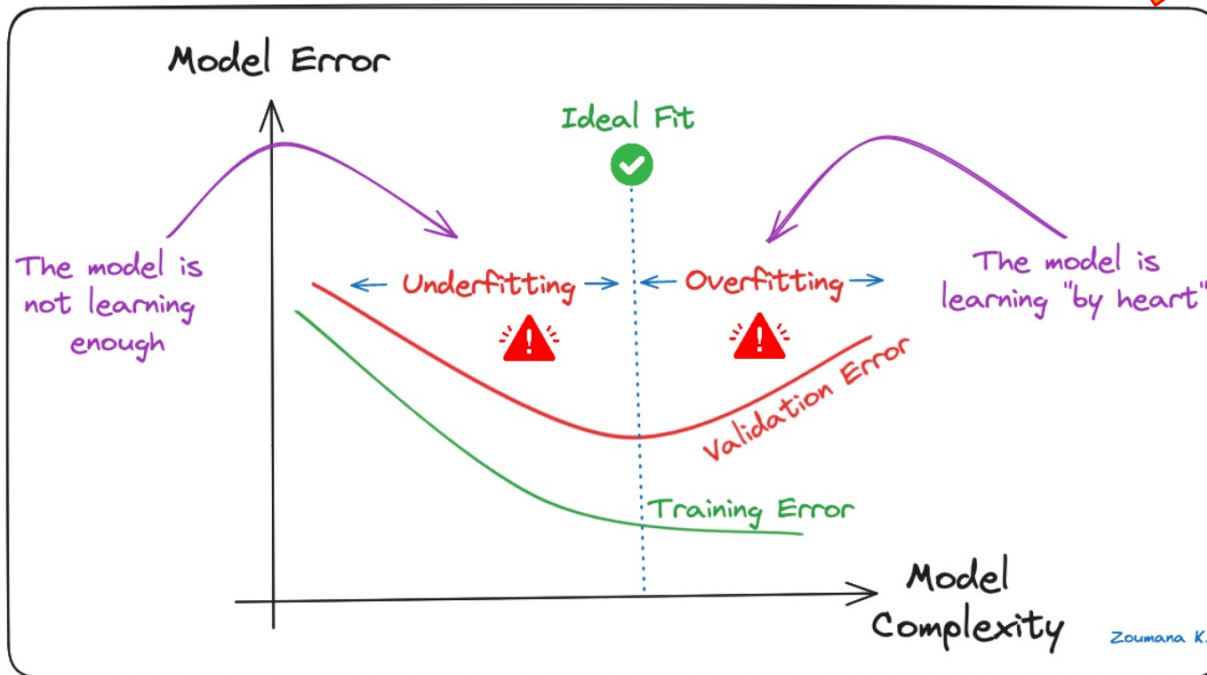
Graphs extracted from: Chen, B., Deng, W., & Du, J. (2017). Noisy Softmax: Improving the Generalization Ability of DCNN via Postponing the Early Softmax Saturation In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5372-5381)



Underfitting Vs. Overfitting

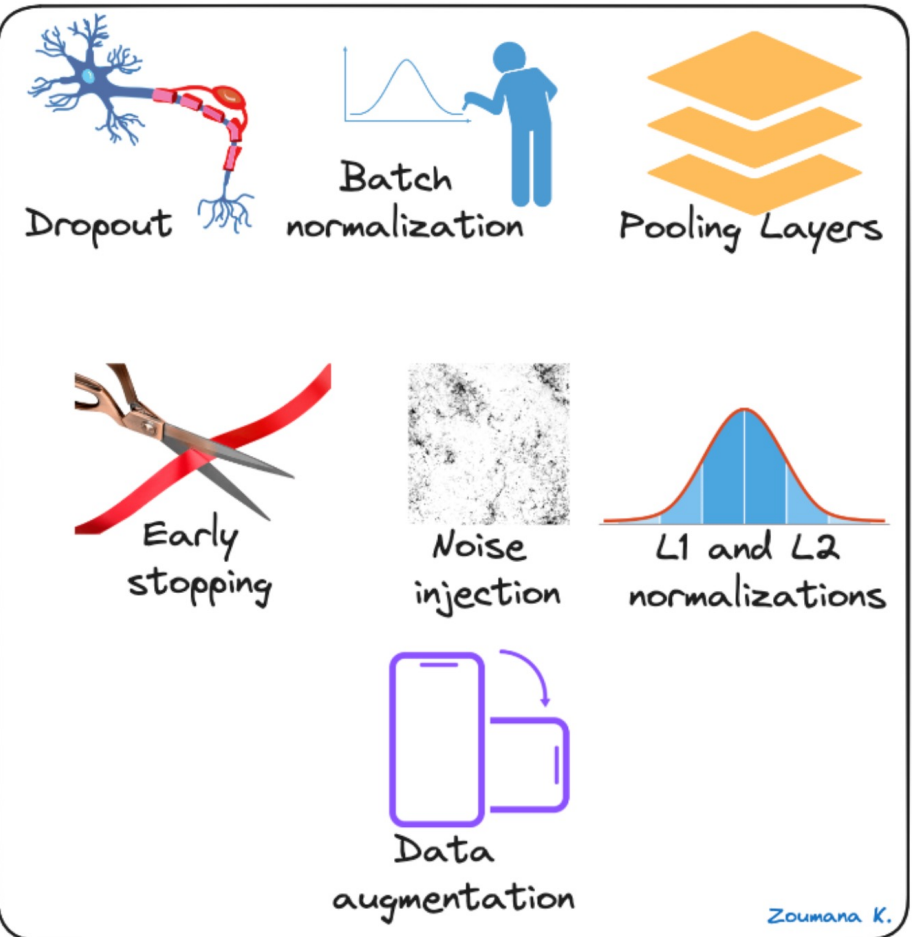
Solutions to take

Underfitting Vs. Overfitting



Infographics extracted from: Keita, Z. (Nov 2023). *An Introduction to Convolutional Neural Network (CNNs)*. <https://www.datacamp.com/tutorial/introduction-to-convolutional-neural-networks-cnns>

7 Strategies to Mitigate Overfitting in CNNs





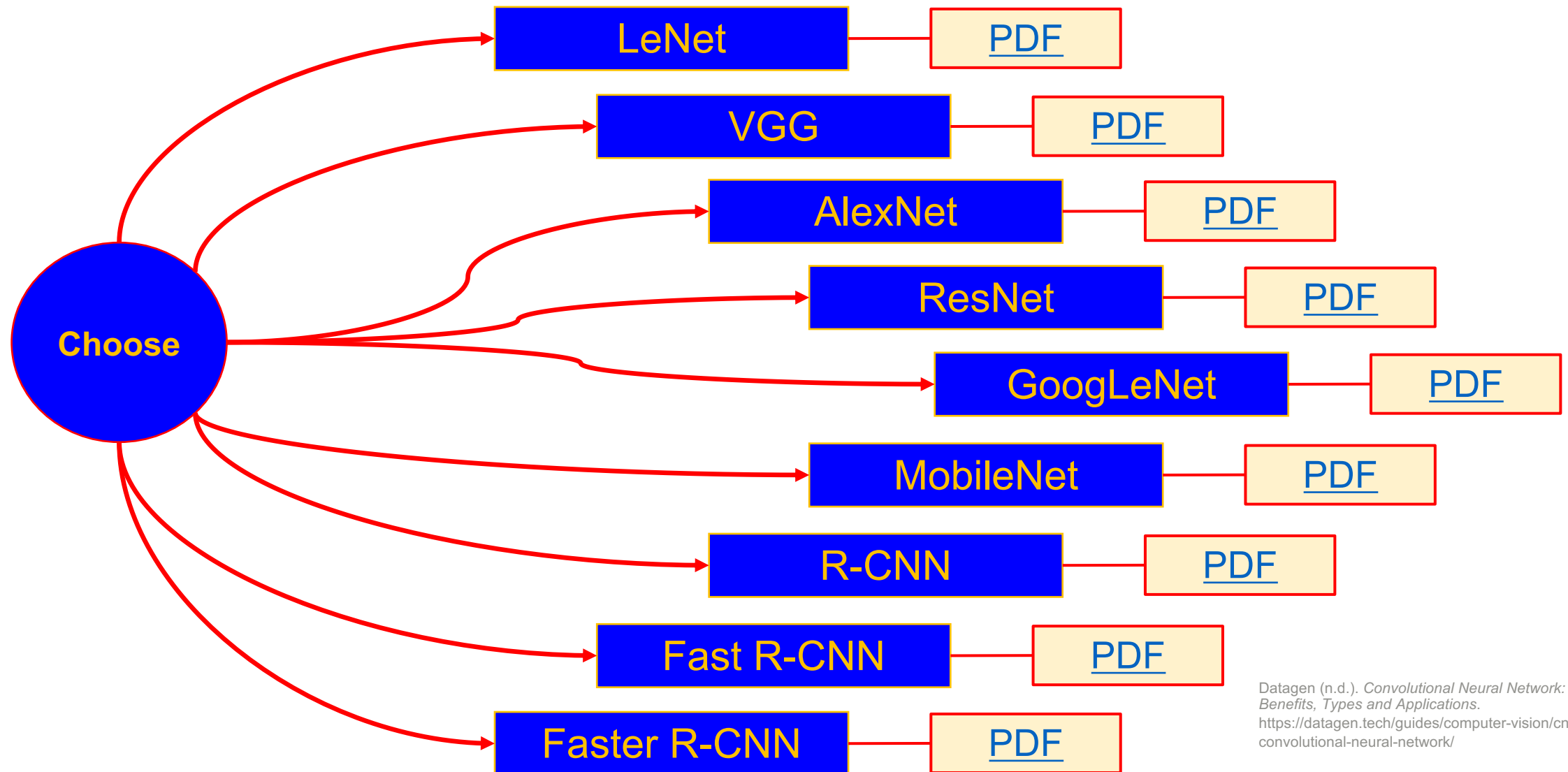
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Applications



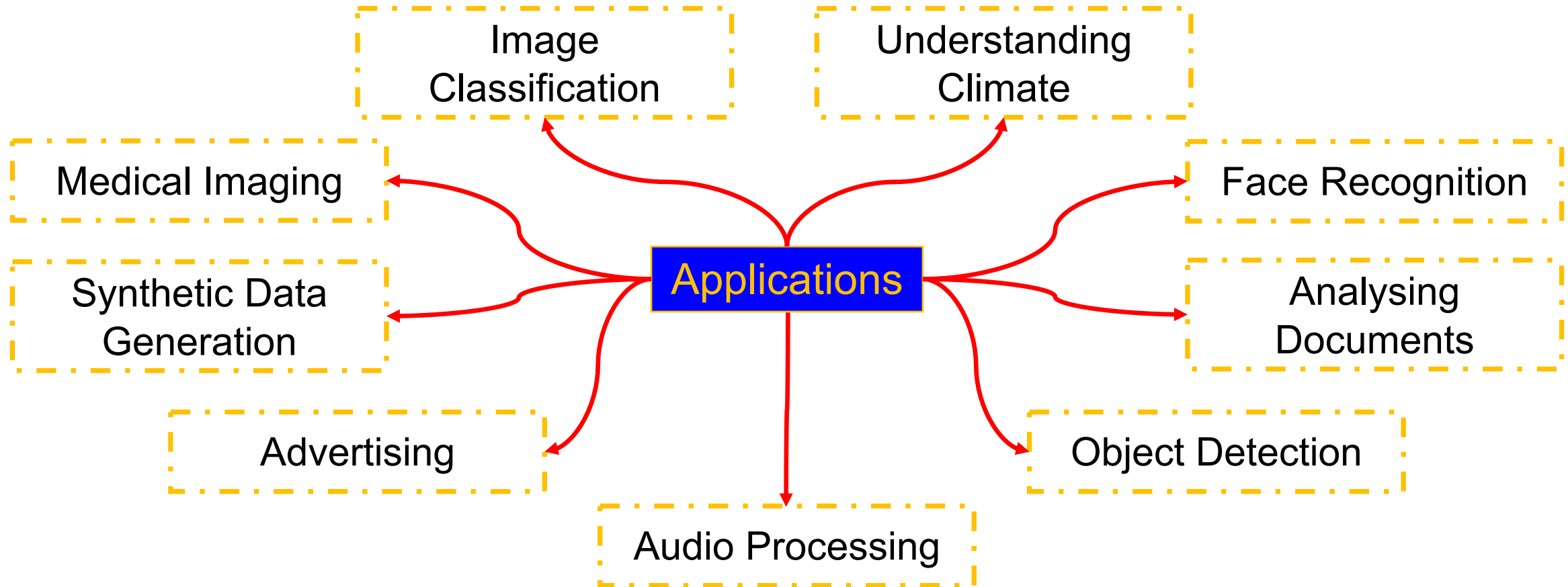
Famous CNN



Datagen (n.d.). *Convolutional Neural Network: Benefits, Types and Applications*.
<https://datagen.tech/guides/computer-vision/cnn-convolutional-neural-network/>



Some Practical Applications of CNN



Keita, Z. (Nov 2023). *An Introduction to Convolutional Neural Network (CNNs)*. <https://www.datacamp.com/tutorial/introduction-to-convolutional-neural-networks-cnns>

Ray, P. (14 Jan 2021). *Convolutional Neural Network (CNN) and its application - All you need to know*. <https://medium.com/analytics-vidhya/convolutional-neural-network-cnn-and-its-application-all-u-need-to-know-f29c1d51b3e5>

MATLAB (n.d.). *What is a Convolutional Neural Network*. <https://www.mathworks.com/discovery/convolutional-neural-network.html>



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Future



Vision Transformer (ViT)

- Designed for Computer Vision with remarkable results
- Uses **neural network** to split images into smaller patches, allowing model(s) to capture both local and global relationships within images.¹

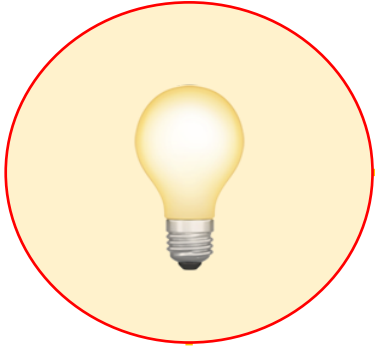


Animation extracted from: Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S. and Uszkoreit, J., (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.

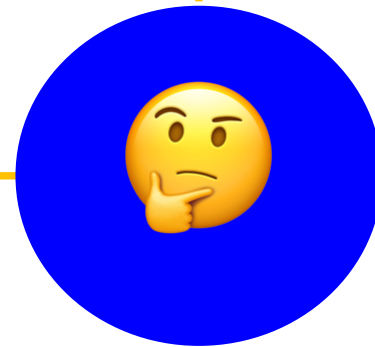
¹**Extracted and modified from:** Hettiarachchi, H. (12 August 2023). *Unveiling Vision Transformers: Revolutionizing Computer Vision Beyond Convolution*. <https://medium.com/@hansahettiarachchi/unveiling-vision-transformers-revolutionizing-computer-vision-beyond-convolution-c410110ef061>



Summary



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Questions



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NEXT:

Image Compression